Specialities of Embedded Systems

Developers must keep in mind these specialties while designing embedded systems.

- 1. Reliability:
- When we use a desktop, sometimes the system 'hangs' and we need to reset the computer. Generally, this does not cause any problem. However, this is not the case with the embedded systems used in mission-critical applications. They must work with high reliability.

- Reliability is of paramount importance in embedded systems. They should continue to work for thousands of hours without break. Many embedded systems used in industrial control are inaccessible. They are hidden in some other large-sized equipment; hence there will not be a reset button on such systems! So, the design of the embedded system should be such that in case the system has to be reset, the reset should be done thematically.
- Special hardware/software needs to be built into the system to take care of it. This special module is known as watchdog timer.
- Many embedded systems used in industrial automation and defence equipment need to work in extreme environmental conditions such as very high/low temperatures, high humidity.
- Besides, they should be able to withstand bump and vibrations. Hence, very stringent environmental specifications have to be met by such systems.
- The ability to work reliably in extreme environmental conditions is known as ruggedness. Not only the military equipment, even the consumer appliances such as the mobile phone need to be very rugged. Many people keep dropping their mobile phone on the floor, but still it works because it is very rugged.

- 2 Performance:
- Many embedded systems have time constraints. For instance, in a process control system, a constraint can be: "if the temperature exceeds 40 degrees, open a valve within 10 milliseconds." The system must meet such deadlines. If the deadlines are missed, it may result in a catastrophe. You can imagine the damage that can be done if such deadlines are not met in a safety system of a nuclear plant.
- 3 Power Consumption:
- Most of the embedded systems operate through a battery. To reduce the battery drain and avoid frequent recharging of the battery, the power consumption of the embedded system has to be very low.
- To reduce power consumption such hardware components should be used that consume less power. Besides, emphasis should be on reducing the components count of the hardware. To reduce component count, the hardware designers have the option of using Programmable Logic Devices (PLDs) and Fie Programmable Gate Arrays (FPGAs). Reducing the component count apart from reducing the pow consumption also increases the reliability of the system.
- 4 Cost: For embedded systems used in safety applications of a nuclear plant or in a spacecraft, cost may be a very important factor. However, for embedded systems used in consumer electronics or of automation, the cost is of utmost importance. Suppose you designed a toy in which the electro will cost US$20. By a careful analysis of the design, if you can find a way to reduce the cost to US it will be a great job. Don't underplay the importance of that one dollar because when you sell million toys, the cost reduction is $10 million! Not surprisingly, the hardware, engineers debate component selection to reduce even $0.1.

- 5. Size:
- Size is certainly a factor for many embedded systems. We do not like a mobile phone that has to carried on our backs. The size and the weight are important parameters in embedded systems use in aircraft, spacecraft, missiles etc. because in such cases, every inch and every gram matters. reduce the size and the weight, again the hardware engineers have to design their boards by reducing the component count to the maximum possible extent. During the initial development of mobile phones based on GSM (Global System for Mobile Communication), GSM was expanded as "God Send Mobiles" because of the complexity of the mobile phone. Today, the mobile phones go into a pockets, thanks to developments in microelectronics.
- 6 Limited User Interface: like desktops, which have full-fledged input/output devices, embedded systems do not have sophisticated interfaces for input and output.
- Some embedded systems do not have any user interface at all.
- They take electrical signals as input and produce electrical signals as output.
- In many embedded stems, the input is through a small function keypad or a set of buttons. The output is displayed either a set of LEDs or a small LCD. For example, the mobile phone has a small display of 4 lines x characters. The input is through a keypad and composing a small text message is not and easy task.Developing a user friendly interface with limitation of the input/output devices is a challenging task firmware developers.

- 7. Software upgradation Capability: Embedded systems are meant for a very specific task. So, once the software is transferred to the embedded system, the same software will run throughout its life. However, in some cases, it may be necessary to upgrade the software.
- Consider the example of a Public Call Office (PCO). At the PCO, an embedded system is used which displays the amount to be paid by a telephone user. The amount is calculated by the firmware, based on the calling number and the duration of the call. From time to time, the telecom operator will change the algorithm for the calculation of the bill amount. So, every time there is a tariff change, the PCO operaror has to replace the program stored in the memory of the embedded system with new program. This is very cumbersome, considering that a memory chip will have to be replaced in thousands of PCOs. Nowadays, software upgradation is done by downloading the software onto the embedded system through a network connection. In future, such an upgradation may become mandatory for many embedded systems.While designing embedded systems, the following aspects need to be considered by the developers: reliability, performance, power consumption, cost, size, limited user interface and software upgradation capability.